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Smartphone Cameras

“Moving Towards SMARTPHONE CAMERAS That Capture The Same QUALITY Photographs As DSLR CAMERAS”

In this selfie era, a smartphone's success depends on the quality of its camera. As a result, manufacturers are putting a lot of focus on digital imaging trends in order to come up with a breakthrough product. *Nidhi Arora* from *Electronics For You*, in an exclusive interview with *Nicolas Touchard*, vice president, marketing, DxOMark, identifies the upcoming innovations in smartphone photography

Q. What are the latest applications driving change in underlying digital imaging technologies?

A. The market for smartphones is highly competitive. Various application utilities like the ability to take high-resolution images in low light, producing ultra-high-definition images, tracking moving objects, improved zoom and Bokeh (depth effect) are driving imaging innovations for smartphone cameras.

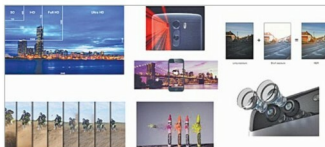
Q. What are the new digital imaging technologies introduced in the last one year? How do they benefit the target applications?

A. In the last one year, we have observed a steep rise in the number of dual-camera phones. This trend will rise in the future as well. We are moving towards smartphone cameras that capture the same quality photographs as DSLR cameras. To achieve this, we need to improve the image capture quality of our phone cameras. In order to do that, cell phone makers have added an extra camera that enables wider and better-quality image capture.

Q. What are the latest standards and frameworks designed to this effect?

A. A smartphone camera consists of lens, sensor and chip. IEEE's TC 42 committee for photonic technology in instrumentation and measurement lays down a standards-based framework for smartphone cameras and communication. It includes members of

working group (WG) 18. Lately, a lot of importance has been given to image stabilisation, sharpness measurement, timing, colours and shadow. The body defines and benchmarks standards for camera communication in scales of shadow and light. Many companies like Apple, Nikon and Google are also representative of the benchmarks. IEEE's CPIQ (Camera Phone Image Quality) goal is to come up with a



Smartphone camera innovations

global evaluation scale that is related to the way consumers look at images.

Q. How does DxOMark help businesses?

A. DxOMark Image Labs provides image-quality testing and optimisation products and services. It provides the industry standard for objectively quantifying camera, lens and mobile image quality. It also provides image quality analysis and consulting services, wherein a pool of experts with professional tools helps the imaging industry at all stages of

camera development and evaluation.

Q. What is the technology behind DxOMark rankings for smartphone cameras?

A. At DxOMark, we test camera quality using various imaging tools, experts and different test situations. Our tests cover a wide spectrum of photography use cases. Over 1500 shots under each environment are analysed along with two videos. This is done in varying standard situations—low light, indoor, outdoor, daylight, fluorescent and more—keeping other factors like time and weather constant. Images are scored based

on 50+ realistic and challenging natural scenes. Scores are produced with the support of scientific tests for exposure, colour, texture, noise, AF, flash, artefacts, zoom and bokeh.

Q. How will these technological advancements impact prices?

A. Higher-end phones are going to cost more. However, in case of dual-camera phones, mid-range segment is going to lead the race. Because of intense market competition, prices won't increase sharply despite better product quality. **EFY**